Talon Flex Fuel Installation and Tuning Plan

1992 Eagle Talon TSi FWD | 4G63 | Stock MHI 14b turbo and stock side mount intercooler

Please convert the car to sensor based flex fuel for pump gasoline e through E85 blends, with automatic fuel compensation in ECMLink v3. Tune for reliable street use within the limits of the 14b and stock intercooler.

Existing equipment

Socketed EPROM ECU with ECMLink v3; AFR500v2 wideband installed (gasoline AFR display). Rebuilt engine with forged internals and Kelford 272/272 cams. Confirm wideband logging and ECU firmware support before installation.

Parts to confirm and install

Item	Specification from build plan
Injectors and resistor bypass	FIC 1440cc High-Z IS126-1440H; PLRESBOX bypass plug.
Pump and regulator	Walbro F90000274 450 LPH E85 pump; WAL-400-1168 install kit; rewire kit. STM Fuelab AFPR kit STMKIT-1GDSM-FPRK-1 with gauge.
Ethanol sensor and mounting	GM/Continental 13507128; Torque Solution TS-UNI-637 dual-pass bracket; two ORB-10 to -6AN fittings (AERO-15609 as listed; verify fit).
Fuel plumbing	E85-rated filter; -6AN PTFE return hose and matching ends; hardline-to-AN adapter sized to the actual chassis return line. Measure hose lengths and fitting angles.
Speed density and wiring	ECMTuning GM MAP/IAT bundle with bung; 1G DSM SD cable; ethanol sensor adapter cable. GM MAP bracket already purchased; select ethanol cable length after routing.

Mechanic installation

- Install injectors with resistor pack bypass, pump with fused relay rewire, AFPR, and ethanol-compatible filter. Inspect existing feed hoses and seals for ethanol compatibility.
- Route return: AFPR → -6AN hose → flex sensor bracket → -6AN hose → factory return hardline. Confirm hardline diameter and adapter type; check regulator control and return capacity with the 450 pump.
- Install MAP/IAT sensors and speed density harness. Once the MAF is no longer used, connect ethanol signal to the ECU MAF input per ECMTuning instructions. Verify sensor pinout, fused switched power, ground, and adapter wiring before energizing. Secure hoses and wiring away from heat and moving parts.

Tuner setup and delivery checks

- Set and record base fuel pressure with reference hose disconnected and plugged, using the tuner's selected pressure. Enter injector calibration for that pressure. Check fuel leaks, boost leaks, sensor readings, and wideband logging.
- Configure speed density and flex fuel; verify ethanol percentage reads correctly. Calibrate gasoline and ethanol endpoints and validate intermediate blends, idle, cruise, cold/hot starts, and boost operation. Use lambda or correctly scaled gasoline-equivalent AFR.
- Log ethanol content, lambda/AFR, knock, timing, boost, injector duty, intake/coolant temperatures, and fuel pressure where instrumented. Confirm fuel delivery under load before increasing boost. Set conservative limits and sensor-fault behavior supported by ECMLink; provide the final tune, logs, fuel requirements, and boost limits.